Mathematical and Computational Methods

Exercise Sheet 9: Integration I: the definite integral and substitution

Tutorial route (▶, in this order, about 80 minutes): 1.1(a), 2.2(a–c), 2.3, 2.4, 2.5, 2.7, 2.9. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Riemann sums and signed area

Exercise 1.1 *(▶ ⋆ drill — a Riemann sum and signed area).* (a) Estimate the area under $f(x) = x^2$ on $[0, 2]$ using $n = 4$ rectangles of equal width and the height at the *right* end of each strip. Sketch the curve with the four rectangles, and say whether your estimate is an over- or under-estimate, and why.

(b) Sketch $g(x) = x - 1$ on $[0, 2]$ and shade the region between the graph and the x-axis, marking the part that counts as *negative* signed area. Find the net signed area $\int_0^2 (x-1)\,\mathrm{d}x$ and the total (unsigned) area.

Indefinite integration: the power rule and the table

Exercise 1.2 *(⋆ drill — rewrite first, then integrate).* Write each integrand as a sum of powers of x and standard forms, then integrate:

$$\text{(a)} \int \left(\sqrt{x} + \frac{1}{x^2}\right) \mathrm{d}x, \qquad \text{(b)} \int \left(2\mathrm{e}^x - \frac{3}{x} + 5\cos x\right) \mathrm{d}x, \qquad \text{(c)} \int \frac{x^2 - 3x + 2}{x}\,\mathrm{d}x.$$

Standard integrals (including arctan/arcsin)

Exercise 1.3 *(⋆ drill — standard forms with a linear inside).* Evaluate the indefinite integrals

$$\text{(a)} \int \mathrm{e}^{3x}\,\mathrm{d}x, \quad \text{(b)} \int \sec^2(2x)\,\mathrm{d}x, \quad \text{(c)} \int \frac{\mathrm{d}x}{9 + x^2}, \quad \text{(d)} \int \frac{\mathrm{d}x}{\sqrt{4 - x^2}}, \quad \text{(e)} \int \cos(5x)\,\mathrm{d}x.$$

FTC Part 2: evaluating definite integrals

Exercise 1.4 *(⋆ drill — definite integrals by FTC 2).* Evaluate

$$\text{(a)} \int_1^3 x^2\,\mathrm{d}x, \qquad \text{(b)} \int_0^{\pi/2} \cos x\,\mathrm{d}x, \qquad \text{(c)} \int_1^{\mathrm{e}} \frac{\mathrm{d}x}{x}, \qquad \text{(d)} \int_0^1 \mathrm{e}^{2x}\,\mathrm{d}x.$$

FTC Part 1: differentiating an integral

Exercise 1.5 *(⋆ drill — FTC Part 1).* Differentiate with respect to x (do *not* try to evaluate the integral):

$$\text{(a)} \frac{\mathrm{d}}{\mathrm{d}x} \int_0^x \mathrm{e}^{-t^2}\,\mathrm{d}t, \qquad \text{(b)} \frac{\mathrm{d}}{\mathrm{d}x} \int_2^x \sin(t^3)\,\mathrm{d}t, \qquad \text{(c)} \frac{\mathrm{d}}{\mathrm{d}x} \int_1^x \frac{\mathrm{d}t}{1 + t^4}.$$

Substitution (reverse chain rule and f'/f)

Exercise 1.6 *(⋆ drill — substitution and the f'/f pattern).* Each integrand contains an "inside" function whose derivative is present. Integrate by substitution; in (c) and (d) the numerator is the derivative of the denominator.

$$\text{(a)} \int 2x\,(x^2+1)^5\,\mathrm{d}x, \quad \text{(b)} \int \cos x\, \mathrm{e}^{\sin x}\,\mathrm{d}x, \quad \text{(c)} \int \frac{2x}{x^2+5}\,\mathrm{d}x, \quad \text{(d)} \int \frac{\mathrm{e}^x}{\mathrm{e}^x+1}\,\mathrm{d}x.$$

Exercise 1.7 *(⋆ drill — changing the limits).* Evaluate $\displaystyle\int_1^2 x\,\mathrm{e}^{x^2}\,\mathrm{d}x$ by changing the limits to the new variable.

No elementary antiderivative

Exercise 1.8 *(⋆ drill — elementary antiderivatives?).* For each function, state whether it has an *elementary* antiderivative.

$$\text{(a) } \mathrm{e}^{-x^2}, \quad \text{(b) } x\,\mathrm{e}^{-x^2}, \quad \text{(c) } \frac{\sin x}{x}, \quad \text{(d) } \sin(x^2), \quad \text{(e) } x\cos(x^2).$$

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core — Riemann sums against the exact area).* Let $f(x) = 1+x^2$ on $[0,3]$, cut into $n = 3$ strips of width 1.

(a) Compute the right-endpoint and left-endpoint Riemann sums.

(b) Using FTC Part 2, find the exact area and check it lies between your two estimates.

(c) Now use n strips of width $\Delta x = \frac{3}{n}$ and right endpoints $x_k = \frac{3k}{n}$. Show that the right-endpoint sum is $S_n = 3 + \dfrac{9(n+1)(2n+1)}{2n^2}$, check that $n = 3$ gives your answer to (a), and find $\lim_{n\to\infty} S_n$. (You may quote $\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1)$.)

Exercise 2.2 *(▶ ⋆⋆ core — FTC evaluation).* *(FTC evaluation — worked-example anchor.)*

(a) Evaluate $\displaystyle\int_1^4 x^2\,\mathrm{d}x$ and state in one sentence what the number represents geometrically.

(b) Evaluate $\displaystyle\int_0^1 \left(3x^2 + \frac{4}{1+x^2}\right)\mathrm{d}x$.

(c) Find the positive value of a for which $\displaystyle\int_0^a (2x+1)\,\mathrm{d}x = 12$.

Exercise 2.3 *(▶ ⋆⋆ core — displacement and distance (physics)).* [Physics / applied] A particle moves on a line with velocity $v(t) = 3t^2 - 12t + 9$ ($\mathrm{m\,s^{-1}}$) for $0 \le t \le 3$.

(a) Find the net displacement $\int_0^3 v\,\mathrm{d}t$.

(b) Find the total distance travelled, $\int_0^3 |v|\,\mathrm{d}t$.

Exercise 2.4 *(▶ ⋆⋆ core — a probability density (data science)).* [Data science / probability] A continuous random variable X has probability density $f(x) = k\,x(1-x)$ on $[0,1]$ and $f(x) = 0$ elsewhere. A density must satisfy $f(x) \ge 0$ and $\int_0^1 f = 1$, and $\mathbb{P}(X \le c) = \int_0^c f$ (probability is area).

(a) Find k, and check that $f \geq 0$.

(b) Find $\mathbb{P}\left(X \leq \frac{1}{2}\right)$ and explain briefly why this value is to be expected.

Exercise 2.5 *(▶ ⋆⋆ core — stating the FTC; FTC 1 with the chain rule).* [Mathematics / pure]

(a) State both parts of the Fundamental Theorem of Calculus, including the hypothesis on f.

(b) Let $g(x) = \int_0^{x^2} \cos t \, \mathrm{d}t$. Find $g'(x)$ in two different ways and check they agree.

Exercise 2.6 *(⋆⋆ core — spot the error).* *(Spot the error / conceptual.)*

(a) A student writes $\int_{-1}^{1} \frac{\mathrm{d}x}{x^2} = \left[-\frac{1}{x}\right]_{-1}^{1} = (-1) - (1) = -2$. The integrand is positive everywhere it is defined, so a negative "area" is impossible. Which theorem did the student use? Check its hypothesis and explain what went wrong.

(b) True or false: "$\int_a^b f(x) \, \mathrm{d}x$ is a function of x." Justify.

Exercise 2.7 *(▶ ⋆⋆ core — substitution with a limit change).* *(Substitution with a limit change — worked-example anchor.)* Evaluate $\int_0^2 x\sqrt{x^2+1} \, \mathrm{d}x$ by converting the limits to the new variable.

Exercise 2.8 *(⋆⋆ core — a stray x to rewrite).* Evaluate $\int_0^3 x\sqrt{x+1} \, \mathrm{d}x$. (Here the substitution leaves a stray x behind — rewrite it in terms of u.)

Exercise 2.9 *(▶ ⋆⋆ core — two routes to a logarithm).* Evaluate two integrals, contrasting the methods:

$$\text{(a) by the } f'/f \text{ pattern, } \int \tan x \, \mathrm{d}x, \qquad \text{(b) by substitution, } \int_1^2 \frac{\ln x}{x} \, \mathrm{d}x.$$

3 Further practice (home)

Properties of the definite integral

Exercise 3.1 *(⋆ drill — properties of the definite integral).* Suppose $\int_0^2 f = 5$ and $\int_2^5 f = -2$. Using only the properties of the definite integral (do *not* find f), evaluate

$$\text{(a) } \int_0^5 f, \quad \text{(b) } \int_5^2 f, \quad \text{(c) } \int_2^2 f, \quad \text{(d) } \int_0^2 3f, \quad \text{(e) } \int_5^0 f.$$

Substitution — indefinite and definite

Exercise 3.2 *(⋆ drill — substitution drill).* Integrate

$$\text{(a) } \int x^2(x^3+1)^7 \, \mathrm{d}x, \ \text{(b) } \int \sin^4 x \cos x \, \mathrm{d}x, \ \text{(c) } \int \frac{(\ln x)^2}{x} \, \mathrm{d}x, \ \text{(d) } \int x\, \mathrm{e}^{-x^2} \, \mathrm{d}x.$$

Exercise 3.3 *(⋆⋆ core — definite substitution).* Evaluate by substitution. In (a) and (c) change the limits; do (b) by *both* routes (return to x, then change the limits) and check that they agree.

$$\text{(a) } \int_0^{\pi/2} \sin x \cos x \, \mathrm{d}x, \qquad \text{(b) } \int_0^1 \frac{x}{(1+x^2)^2} \, \mathrm{d}x, \qquad \text{(c) } \int_0^1 \frac{2x}{x^2+1} \, \mathrm{d}x.$$

arctan **and** arcsin **targets**

Exercise 3.4 *(⋆⋆ core — inverse-trig targets and completing the square).* Evaluate

$$\text{(a)} \int_0^{\sqrt{3}} \frac{\mathrm{d}x}{1+x^2}, \quad \text{(b)} \int_0^{1} \frac{\mathrm{d}x}{\sqrt{4-x^2}}, \quad \text{(c)} \int_0^{2} \frac{\mathrm{d}x}{4+x^2}, \quad \text{(d)} \int \frac{\mathrm{d}x}{x^2+2x+2}.$$

In (d), complete the square first.

Signed area versus total area

Exercise 3.5 *(⋆⋆ core — signed area and total area).* For each function find both the net signed area and the total (painted) area over the stated interval.

(a) $\sin x$ on $[0, 2\pi]$.

(b) $x^2 - 1$ on $[0, 2]$.

Symmetry

Exercise 3.6 *(⋆⋆ core — odd and even on a symmetric interval (maths)).* [Mathematics / pure] Let $a > 0$ and let f be continuous on $[-a, a]$. Using the substitution $u = -x$ on $\int_{-a}^{0} f$, prove that $\int_{-a}^{a} f(x)\,\mathrm{d}x = 0$ whenever f is *odd*, and that it equals $2\int_0^a f(x)\,\mathrm{d}x$ whenever f is *even*.

Challenges and extensions

Exercise 3.7 *(⋆⋆⋆ challenge — FTC 1 with the chain rule).* Use FTC Part 1 together with the chain rule to differentiate

$$\text{(a)} \frac{\mathrm{d}}{\mathrm{d}x}\int_0^{x^2} \mathrm{e}^{t}\,\mathrm{d}t, \qquad \text{(b)} \frac{\mathrm{d}}{\mathrm{d}x}\int_x^{x^2} \ln\!\left(1+t^2\right)\,\mathrm{d}t.$$

Exercise 3.8 *(⋆ ⋆ ⋆ challenge — completing the square, log plus arctan).* A *linear* numerator over an irreducible quadratic splits into a logarithm plus an arctangent. Evaluate $\displaystyle\int \frac{x+1}{x^2+4x+5}\,\mathrm{d}x$. *(Hint: complete the square below, and write the numerator as a multiple of the denominator's derivative plus a constant.)*

Exercise 3.9 *(⋆⋆⋆ challenge — no elementary antiderivative (data science)).* [Data science / numerical] Consider $I = \displaystyle\int_0^1 \mathrm{e}^{-x^2}\,\mathrm{d}x$ and $J = \displaystyle\int_0^1 x\,\mathrm{e}^{-x^2}\,\mathrm{d}x$.

(a) Exactly one of them can be evaluated by FTC 2 with an elementary antiderivative. Which one? Evaluate it, and explain why FTC 2 cannot evaluate the other in elementary terms.

(b) Bracket the other integral using left- and right-endpoint Riemann sums with $n = 2$ ($\Delta x = \frac{1}{2}$). Its true value is ≈ 0.7468; is it inside your bracket?